

Homework

No Teams (every student must make an individual submission)

- (1) Start by writing a parser (C,C++, compiles with make) on Ubuntu for the MovieLens Dataset (use the 10M data set)
 - <https://press.liacs.nl/movielens/10M/>
 - *tip: you probably want to load the dataset into RAM memory*

After writing the parser, then write a command line program called moviesearch which allows simple text searches for movies by keywords in the title, year, genre and tag

Command: **moviesearch –title [keywords] –year [year] –genre [genres] –tag [tags]**

Where all information given in the query is required (an AND search) – If you specify 2 keywords then both must be in the title; If you specify multiple genres, then all must be in the genre list.

Example Query 1: **moviesearch –title Blood –tag Upton**
Would give
56782::There Will Be Blood (2007)::Drama

Example Query 2: **moviesearch –title Las Vegas**
Would give
25::Leaving Las Vegas (1995)::Drama|Romance
1884::Fear and Loathing in Las Vegas (1998)::Adventure|Comedy|Crime|Drama|Fantasy
7420::Viva Las Vegas (1964)::Comedy|Musical|Romance
 - (2) A screenshot for each example query: (1) **moviesearch –title Blood –tag Upton** (2) **moviesearch –title Las Vegas**
 - Put the answers in a zip file of a directory called your name (e.g. joost.kok, with subdirs P1 and P2, one subdir per problem)
 - A directory, code, which contains your sourcecode with Makefile(s) (where appropriate)
 - A file, answers.txt which has your assessment if you solved each problem and also specifies the LIACS computer room and computer name (see side of computer) where it worked
 - **Do not put the MovieLens 10M dataset in the zip file** – e.g. Your program should assume that the original *.dat files are in the same directory as the executable (e.g. fp = fopen("ratings.dat", "r");)
 - *joost.kok*
 - P1
 - code
 - answers.txt
 - P2
 - Screenshot for Example Query 1
 - Screenshot for Example Query 2
- Due date: see LCM (roughly one week, late penalties as mentioned in the slides)